



# Training Python-Django - 0

## Initiation

*Summary: This first day will allow you to familiarize yourself with the basics of web development. Here's what's on the menu: HTTP, HTML, CSS and integration of existing Javascript scripts into your pages.*

*Version: 1.1*

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# Chapter I

## Preamble

Here is what Wikipedia has to say about *Balaenoptera musculus*:

The blue whale (*Balaenoptera musculus*) is a marine mammal belonging to the baleen whale suborder Mysticeti. Reaching a maximum confirmed length of 29.9 meters (98 feet) and weight of 177 tonnes (190 tons), it is the largest animal known to have ever existed.

There are currently five subspecies of blue whale, recognized by the Society for Marine Mammalogy's Committee on Taxonomy: *B. m. musculus* in the North Atlantic and North Pacific, *B. m. intermedia* in the Southern Ocean, *B. m. brevicauda* (the pygmy blue whale) in the Indian Ocean and South Pacific Ocean, *B. m. indica* in the Northern Indian Ocean, and *B. m. unnamed subsp.* in the waters off Chile. The blue whale's diet consists almost exclusively of euphausiids (krill).

Blue whales were abundant in nearly all the oceans on Earth until the end of the 19th century. They were hunted almost to extinction by whaling until the International Whaling Commission banned all hunting of blue whales in 1967. The International Whaling Commission catch database estimates that 382,595 blue whales were caught between 1868 and 1978. The global blue whale population abundance is estimated to be 10,000–25,000 blue whales, roughly 3–11% of the population size estimated in 1911. There remain only much smaller concentrations in the Eastern North Pacific (1,647), Central North Pacific (63–133), North Atlantic (1,000–2,000), Antarctic (2,280), New Zealand (718), Northern Indian Ocean (270), and Chile (570–760). It is considered an endangered species.

No whale was harmed during the writing of this subject.

# Chapter II

## General rules

- Your project must be realized in a virtual machine.
- Your virtual machine must have all the necessary software to complete your project. These softwares must be configured and installed.
- You can choose the operating system to use for your virtual machine.
- You must be able to use your virtual machine from a cluster computer.
- You must use a shared folder between your virtual machine and your host machine.
- During your evaluations you will use this folder to share with your repository.
- Your functions should not quit unexpectedly (segmentation fault, bus error, double free, etc) apart from undefined behaviors. If this happens, your project will be considered non functional and will receive a 0 during the evaluation.
- We encourage you to create test programs for your project even though this work **won't have to be submitted and won't be graded**. It will give you a chance to easily test your work and your peers' work. You will find those tests especially useful during your defence. Indeed, during defence, you are free to use your tests and/or the tests of the peer you are evaluating.
- Submit your work to your assigned git repository. Only the work in the git repository will be graded. If Deepthought is assigned to grade your work, it will be done after your peer-evaluations. If an error happens in any section of your work during Deepthought's grading, the evaluation will stop.



For obvious security reasons, any credentials, API keys, env variables etc... must be saved locally in a `.env` file and ignored by git. Publicly stored credentials will lead you directly to a failure of the project.

# Chapter III

## Exercise 00

|                                                                                   |            |
|-----------------------------------------------------------------------------------|------------|
|  | Exercise00 |
| Exercise 00: First shell script                                                   |            |
| Directory: <i>ex00/</i>                                                           |            |
| Files to Submit: <code>myawesomescript.sh</code>                                  |            |
| Authorized: <code>curl</code> , <code>grep</code> , <code>cut</code>              |            |

If Twitter has no secrets for you, you probably know [bit.ly](https://bit.ly): a very useful URL shortening service.

The goal of this exercise is to write and turn in a shell script that displays the real address of a supposedly valid `bit.ly` address (that is, "the address the `bit.ly` link leads to").

As stated in this exercise header, you can only use the following shell commands: `curl`, `grep` and `cut`. Your best bet is to start reading the `curl` manual. To do so, type `man curl` in your terminal.

Here is an example of how your shell script should behave:

```
$> ./myawesomescript.sh bit.ly/1072s3U
http://42.fr/
$>
```

The example above clearly shows your script must be executable. You must use `/bin/sh` as an interpreter.

Turn in your script in the `ex00` folder at the root of your repo.

# Chapter IV

## Exercise 01

|                                                                                   |            |
|-----------------------------------------------------------------------------------|------------|
|  | Exercise01 |
| Exercise 01: Your resume in HTML                                                  |            |
| Directory: <code>ex01/</code>                                                     |            |
| Files to Submit: <code>cv.html</code>                                             |            |
| Authorized: <code>n/a</code>                                                      |            |

You will write your resume in HTML/CSS respecting the following constraints:

- You must respect the semantics of your HTML tags, as well as the separation between style and content.
- You must create a consistent HTML file with the minimum required content: name, surname, skills and career path.
- You must display at least one title with the `title` tag and a title with the `h1` tag.
- You must use at least one table with the `table`, `th`, `tr` and `td` tags.
- You must use at least a list with the `ul` tag and a list with the `ol` tag. The elements must use a `li` tag.
- The table borders must be visible (`solid`). The table borders must be merged (`collapse`).
- The lower-right cell of a table must have a `#424242` border color.
- You must use a different syntactic solution for each previous instruction: for the first one, use the `style` tag in the `head` of your page. For the second, use a `style` attribute in the tag you see fit.



No special instruction about the veracity of information. You can craft a crazy resume if you like, as long as you follow the instructions above.

# Chapter V

## Exercise 02

|                                                                                   |            |
|-----------------------------------------------------------------------------------|------------|
|  | Exercise02 |
| Exercise 02: Email sending form                                                   |            |
| Directory: <i>ex02/</i>                                                           |            |
| Files to Submit: <b>form.html</b>                                                 |            |
| Authorized: <b>n/a</b>                                                            |            |

Create a HTML form that represents the usual information of any contact. This form will show the following fields:

- **Firstname:** a text field.
- **Name:** a text field, also.
- **Age:** you must use the specific numeric field specific to HTML5.
- **Phone:** you must use the tel field specific to HTML5.
- **Email:** you must use the email field specific to HTML5.
- **Student at 42?:** you must use the checkbox field.
- **Gender:** you must use radio buttons with the values **Male**, **Female** and **Other**.
- A form submission button. The **onclick** attribute of your button must be: **displayFormContents**

The tarball `d00.tar.gz` in this subject appendix contains an `ex02/` sub-folder that contains a JavaScript `popup.js` file written by your boss's son, who's an intern in your company. And since you would not like to make your boss's son feel like an incompetent slob as far as programming goes, you cannot modify his file, which must be used as is.



A thorough reading and a superficial understanding of the provided JavaScript code are required to complete this exercise.

You must correctly integrate this JavaScript file in your HTML page. If your HTML code is correct, pushing the form button will make a super modern popup appear. It will contain the fields and values of your form. If it doesn't, your HTML code is flawed.

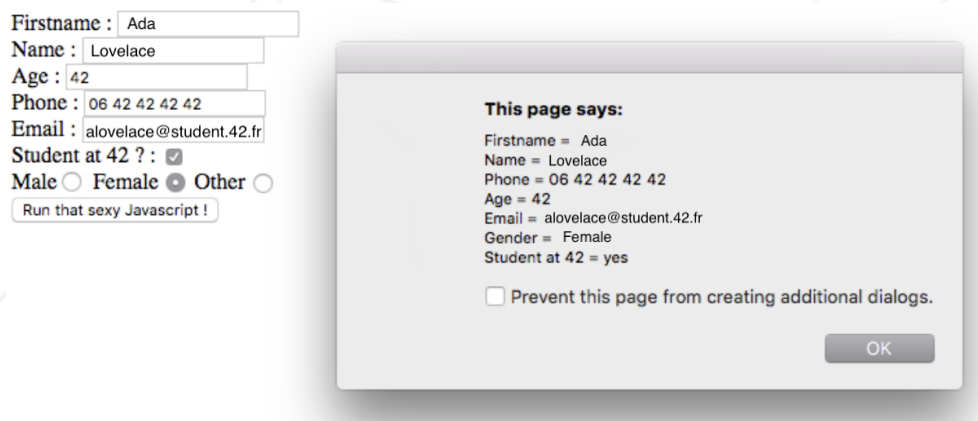


Figure V.1: Non-contractual illustration of the expected result.



# Chapter VI

## Exercise 03

|                                                                                   |            |
|-----------------------------------------------------------------------------------|------------|
|  | Exercise03 |
| Exercise 03: Web page replication                                                 |            |
| Directory: <i>ex03/</i>                                                           |            |
| Files to Submit: <code>copy.html</code>                                           |            |
| Authorized: <code>n/a</code>                                                      |            |

A competing business has uploaded a website that's nicer than yours. Thanks to a serious mission of industrial espionage, your boss gets a screenshot of a page and its CSS file. You can access both those files in the appendix of this subject in the `d00.tar.gz` archive and its `ex03/` sub-folder.

You must replicate this page as faithfully as possible!

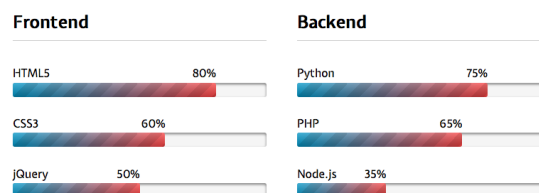


Figure VI.1: The screenshot of the page you must replicate. Image scale is non-contractual.

Once again, you will have to separate style and content, respect the tags' semantics you will use and keep the structure logical in your document.

You must use the provided CSS file without modifying it. A "fresh" version of the CSS will be used during the evaluation in order to check whether you have followed this instruction.

# Chapter VII

## Exercise 04

|                                                                                   |            |
|-----------------------------------------------------------------------------------|------------|
|  | Exercise04 |
| Exercise 04: Snippets JS integration                                              |            |
| Directory: <i>ex04/</i>                                                           |            |
| Files to Submit: <b>snippets.html</b>                                             |            |
| Authorized: n/a                                                                   |            |

The `d00.tar.gz` tarball in this subject appendix contains an `ex04/` sub-folder that contains the same 4 files: `file1.js`, `file2.js`, `file3.js` and `file4.js`.

You must create and turn in a **snippets.html** file that must import the 4 scripts so that the pop-up appears **correctly** (meaning no strange characters should appear).



You must import the specified scripts. You cannot modify them. You cannot add JavaScript in your HTML code.

# Chapter VIII

## Exercise 05

|                                                                                   |            |
|-----------------------------------------------------------------------------------|------------|
|  | Exercise05 |
| Exercise 05: W3C validation                                                       |            |
| Directory: <i>ex05/</i>                                                           |            |
| Files to Submit: Your edited <code>index.html</code>                              |            |
| Authorized:                                                                       |            |

Code is nice. Nice code is better. And to write nice code, you should follow a nice norm.

The **W3C standard** is a staple, and you have to respect its form when writing or generating HTML.

In the `d00.tar.gz` tarball located in this subject appendix you will find the `ex05/` sub-folder. It contains the sources of a complete web page. Unfortunately, it was written by a developer far less skilled than yourself!

Edit the HTML code of the `html` index file so it can pass the [W3C validation](#)! This means neither errors nor *warnings*.

You must *edit* the file, not truncate it. This means the file's content you will edit *must* be included in its entirety in your repo.

# Chapter IX

## Submission and peer-evaluation

Turn in your assignment in your `Git` repository as usual. Only the work inside your repository will be evaluated during the defense. Don't hesitate to double-check the names of your folders and files to ensure they are correct.



The evaluation process will happen on the computer of the evaluated group.